

A 100-g block of iron at 31°C is submerged in 50 g of water at 25°C in a thermally insulated container. If the temperature of the system reaches equilibrium at 26°C, what is the specific heat of iron? (Note: The specific heat of water is 4.2 J/(g·°C).)

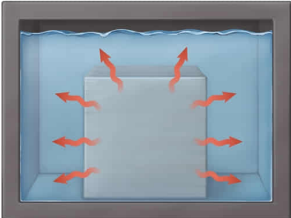
- ✓ ☒ A. 0.42 J/(g·°C) [60%]
☐ B. 1.7 J/(g·°C) [23%]
☐ C. 42 J/(g·°C) [12%]
☐ D. 170 J/(g·°C) [3%]

Correct

60% answered correctly

Explanation:

$q = mc\Delta T$



Heat gained by water

 $q_{\text{water}} = (50 \text{ g}) \left(4.2 \frac{\text{J}}{\text{g}\cdot^\circ\text{C}} \right) (1^\circ\text{C})$

Heat lost by iron

 $q_{\text{iron}} = (100 \text{ g}) (c_{\text{iron}}) (-5^\circ\text{C})$

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The specific heat of a substance is an intrinsic property (does not depend on mass) and is the amount of heat required to increase a unit mass (1 g or 1 kg) by 1°C or 1 K. The total amount of **heat gained or lost** by an object is the product of its mass m , its specific heat c , and the change in temperature ΔT , as given by the **calorimetry equation**:

$$q = m c \Delta T$$

Because the block of iron is hotter than the water, heat flows from the iron to the water, and the temperature of the water increases from 25°C to 26°C ($\Delta T = 1^\circ\text{C}$). The heat gained by water is calculated using its given mass (50 g), specific heat $\left(4.2 \frac{\text{J}}{\text{g}\cdot^\circ\text{C}} \right)$, and ΔT :

$$q_{\text{water}} =$$

Because the container is thermally insulated, the heat gained by the water is equal to the heat lost by the iron block, $q_{\text{iron}} = -210 \text{ J}$. According to the question, the 100-g iron block decreases in temperature from 31°C to 26°C ($\Delta T = -5^\circ\text{C}$). The **specific heat** of iron is calculated by rearranging the calorimetry equation and using these values:

$$c_{\text{iron}} =$$

(Choice B) The amount of heat transferred depends on the change in temperature ΔT , not the initial temperature of the substance. If the initial temperatures of iron (31°C) and water (25°C) are incorrectly used, $c_{\text{iron}} = 1.7 \text{ J/(g}\cdot^\circ\text{C)}$.

(Choice C) Whereas the *specific heat* (c) of a substance is an intrinsic property, its *heat capacity* (C) is the amount of heat required to raise a given amount of the substance (mass) by 1°C. The heat capacity of the iron block is the product of specific heat and mass: $C = \left(0.42 \frac{\text{J}}{\text{g}\cdot^\circ\text{C}} \right) (100 \text{ g}) = 42 \frac{\text{J}}{^\circ\text{C}}$.

(Choice D) If the *heat capacity* of iron is incorrectly determined by using the initial temperatures of the iron block and water, it is calculated as 170 J/°C.

Educational objective:

The amount of heat q gained or lost by an object is the product of its mass m , its specific heat c , and its change in temperature ΔT : $q = mc\Delta T$, where specific heat is the amount of heat required to increase a unit mass of a substance by 1°C (or 1 K).

Time spent: 0 sec
 Subject:
 Physics

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